

WOC7015 Group Project Description

University/Programme/Course: Universiti Malaya / Master of Software Engineering /Software Verification and Validation

Pedagogical Approach: Project-based Learning

Mediums: Spectrum, Telegram, and MS Teams.

Learning Outcomes:

1. Prepare a full coverage Test Plan with good strategies.
2. Professional communication between team members.

This projek aims to prepare a Test Report focusing on Test Plan and conduct tests (the number of tests depending on your test plan) for an open source software to achieve full test coverage.

The open source software (assigned to you) requirements specification has been given to you through the [link](#).

The specification helps you to understand in general on how the open source software works. You need to download and execute the software to allow you to test it through the strategy, approaches, and techniques you planned in the Test Report. If you are unable to test according to the plan due to infrastructure or technicality issue, please explain them in the report.

To help you with this assessment, you may refer to [Expense Manager case study](#) with its [test report](#). Please follow the format. However, please be reminded that the test report is incomplete and partially accurate. It is meant for your reference only.

Please make sure each member indicates his/her name on the work done.

Scope

To meet the assessment requirements, you will need to:

- Form a group of five (5) members. One member will be responsible as a group leader. Read and review the open source software brief description assigned to you. You must download and run the software on your machine to understand how the software works. The specification may be incomplete and not thorough enough to deliver all software requirements. Each member tests at least two (2) functional requirements and one (1) non-functional requirement.
- The **test report** consists of Test Design with Test Case Specification, Test Procedure Specification, Model Checking, User Acceptance, Test Log, Test Summary Report, and Test Completion Report.

- **Use Case Testing** technique is compulsory to design and use other than your proposed testing techniques. You have to carefully design tests for non-functional requirements. You may look into the standard or regulation involved in executing the software. The chosen techniques will determine how much coverage of the program code you will conduct. Please be careful on the risk value you will give for each functionality.
- You must provide explanation on your **Approach Refinement** to justify of the techniques you have chosen. The total number of techniques for each group is at least 10 techniques. You must indicate the test level. The mark given on how extensive tests you will conduct on all aspects.
- There are four state diagrams need to be checked using Model Checking. You must justify the **Model Checking** technique you used for each state diagram.
- Conduct the tests and record the result on the **Test Log**. The more defects you find, the more marks the group will get. However, each member will be assessed based on how extensive the test coverage based on techniques he/she chosen. Hence, each member will have different mark. The group leader is responsible of the compilation of the report and the report's consistency, therefore, he/she will get extra 7 marks.
- Develop user acceptance testing using an established questionnaire model. You must conduct the user acceptance testing using not less than 30 respondents. You must generate graphs to visualize the result of the survey. You are required to produce a **User Acceptance Test Report**.
- Write a summary of the test report produced with recommendations for the software development team on the **Test Summary Report** and **Test Completion Report**. The recommendations include how the software can be improved to fit with the current situations.

Assessment

The distribution of marks for the Test Report is described in Table 1.

Table 1: Distribution of Marks

Assessment Components
1. Formatting, consistency, and completeness of Test Report (10%) 2. Test Design 25% a. Completeness – 5% b. Approach – 20% 3. Test Case – 15% 4. Test Procedure Specification – 5% 5. Model Checking Report – 10% 6. User Acceptance Report – 10% 7. Test Log – 10% 8. Test Summary Report – 10% 9. Test Completion Report – 5%
Total : 100%

Submission Guideline:

- 1- This is a group alternative assessment. ONLY GROUP LEADER SUBMITS THE REPORT.
- 2- You must use this file name format for the submission –
WOC7015-AA-YourName.fileFormat

For example: WOC7015-AA-SitiHafizah.docx
- 3- You must submit two versions of file format (docx and pdf). Please zip your submission.
- 4- Please submit to the Spectrum under Alternative Assessment (19 January, 2024 @ 11.55pm)” link.
- 5- Submission policy:
 - a. One (1) mark will be deducted for every ~~within~~ 30 minutes late.
 - b. Five (5) marks will be deducted for wrong format submission.
 - c. Five (5) marks will be deducted for wrong file name used.

**TAMAT
END**